

SANYA KAPOOR

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PROFILE SUMMARY

Data Science postgraduate with experience in machine learning, statistical modelling, experimentation and large-scale data analysis, as well as business intelligence and reporting, to solve real-world problems. Skilled in Python, SQL, and AWS, with experience building predictive models, analysing complex datasets, and translating insights into actionable recommendations through academic and industry projects. Strong foundations in statistics, forecasting, and hypothesis testing with experience in communicating findings to technical and non-technical stakeholders. Open to relocation and eligible to work in the UK.

TECHNICAL SKILLS

Programming & Databases: Python (Pandas, NumPy, Scikit-learn, Matplotlib), SQL, R

Machine Learning: Supervised Learning, Classification, Feature Engineering, Model Evaluation, Cross Validation, Time Series Forecasting (Prophet, ARIMA)

Statistics & Experimentation: Statistical Modelling, Hypothesis Testing, A/B Testing, Predictive Analytics, Data Interpretation

Data Engineering: ETL Pipelines, Data Integration, Data Cleaning

Visualisation & BI: Power BI, Tableau, Excel, KPI Reporting

Cloud & Automation: Git, AWS, Terraform Cloud

Deployment: Streamlit

EDUCATION

University of Bristol, MSc Data Science

Bristol, United Kingdom | Sept 2025 – 2026

Relevant Coursework: Statistical Computing and Empirical Modelling, Visual Analytics, Large-Scale Data Engineering, Machine Learning, AI and Text Analytics.

SRM Institute of Science & Technology, B.Tech. Electronics & Communication

Chennai, India | 2019 – 2023

GPA: 9.21/10

WORK EXPERIENCE

Tata Communications — Engineer, Global IP & Ethernet

Chennai, India | July 2023 – March 2025

- Automated network monitoring workflows using Python and REST APIs by replacing manual checks across enterprise-scale IP and Ethernet infrastructure.
- Supported enterprise-scale network operations and customer network solutions across routing, switching, and AWS cloud services.
- Collaborated with cross-functional engineering teams to implement customer network solutions and resolve operational issues.

Mentor Minds — Data Analyst Intern

India | May 2023 – June 2023

- Built weekly Power BI reports tracking eight key business performance metrics for stakeholder review and translated requirements into KPI dashboards with drill-down functionality.
- Built SQL & Python ETL pipelines integrating customer behaviour data from three data sources and conducted quantitative and qualitative analysis to identify user behaviour trends and inform business improvement recommendations.

PROJECTS & RESEARCH

Forecasting Physical Activity Participation In London | Dissertation with London Sport

- Harmonised and analysed 135K+ responses from eight waves of the Sports England Active Lives Survey across 32 London boroughs, built demographics, behavioural and geographic attributes for modelling.
- Developed and compared Ordinal Logistic Regression, LightGBM, CatBoost, K-Modes clustering & Bayesian Hierarchical models to model physical activity across Inactive, Fairly Active and Active categories.
- Built a forecasting framework which combines activity propensity with projected demographic composition and behavioural trends, used rolling-origin validation and unseen-wave testing to assess generalisation.
- Achieved 0.704 AUC with the final ordinal logistic model and the forecasting framework achieved 2.74pp mean MAE against 3.44pp for the persistence baseline.

Customer Retention Prediction System

- Built an end-to-end machine learning pipeline to predict customer churn for a banking dataset & addressed class imbalance with SMOTE.
- Translated model output into targeted retention recommendations and deployed findings as an interactive Streamlit application for exploring churn risk by segment.

UK Housing Price Analysis

- Built a predictive analytics pipeline on UK housing price data, from data cleaning and exploratory analysis through feature engineering.
- Delivered results through an interactive Tableau dashboard to explore regional trends and compare models and found substantial regional disparities with London prices nearly 3x higher than other UK regions.

Socio-Economic Housing Inequalities in England & Wales

- Analysed UK Census data covering 7,200+ MSOAs to investigate regional housing inequality patterns, applied dimensionality reduction and statistical techniques to identify socio-economic trends across regions.
- Designed Tableau dashboards that communicated analytical findings to end users.

Renal Disease Prediction Using Machine Learning and Medical Imaging Approaches.

[IEEE Xplore Publication](#)

- Built an XGBoost classification model predicting kidney disease status from clinical parameters, supported by correlation analysis and class-distribution EDA and spanned it across CT scan segmentation and a daily health-tracking application.

CERTIFICATIONS

AWS Certified Cloud Practitioner | Microsoft Azure AI Fundamentals | Databricks Generative AI Fundamentals | Databricks Fundamentals Accreditation | Big Data Analytics | SQL for Data Science